

# Resin Pouring — Production Logging

Operations Handbook · For management

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An hourly production record for the pouring station, kept by the operators on the phones they already carry. This document is written for whoever decides whether the plant runs it: what it holds, what it answers, what it costs the floor, and what else it could do. Part One is what runs today. Part Two is what the same record can be asked to do, each of which is a setting rather than a requirement. **The floor has its own document** — a separate operator guide covering a shift start to finish — so nothing here is written for an operator to follow.

# What is in this handbook

Part One is the system as it runs: one screen an operator uses, and everything read back out of what they enter. **Nothing in it requires an entry from anyone but the operator.** Part Two is what the same record supports if it is wanted, each one a setting that is switched off until somebody switches it on.

## PART ONE – IN SERVICE

|    |                                            |                                                   |    |
|----|--------------------------------------------|---------------------------------------------------|----|
| 01 | <b>What it records</b>                     | What the record holds, and what it asks of anyone | 3  |
| 02 | <b>How it fits together</b>                | One database, and every screen a window onto it   | 5  |
| 03 | <b>The operator terminal</b>               | The one screen an operator uses                   | 6  |
| 04 | <b>Lot traceability</b>                    | Which lot went into which containers              | 7  |
| 05 | <b>What comes out of it</b>                | Scrap, trends and getting the record out          | 8  |
| 06 | <b>The floor as it is now</b>              | The live view of the floor as it stands           | 11 |
| 07 | <b>Who can do what</b>                     | Roles, and what each one can reach                | 12 |
| 08 | <b>Running it</b>                          | Backups, accounts, and keeping the record honest  | 13 |
| 09 | <b>Vessel levels and the floor display</b> | What is left in each vessel, and the wall display | 15 |

## PART TWO – AVAILABLE IF WANTED

|    |                                   |                                                |    |
|----|-----------------------------------|------------------------------------------------|----|
| 10 | <b>Assigning work to stations</b> | Dispatching runs to stations and tracking them | 17 |
| 11 | <b>Fill weight and giveaway</b>   | Recording what a container actually weighed    | 18 |

## APPENDICES

|   |                                        |                                                            |    |
|---|----------------------------------------|------------------------------------------------------------|----|
| A | <b>Guide — Manager</b>                 | Working guides for the manager and the administrator       | 20 |
| B | <b>Reading the screen on the floor</b> | Gloves, lighting, contrast and colour on the floor         | 23 |
| C | <b>How it is tested</b>                | What is checked before a release, and what that has caught | 24 |
| D | <b>Built, not in service</b>           | Built, not in use, and not proposed                        | 25 |

### WHO THIS IS FOR, AND HOW TO READ IT

Management. The working guides in Appendix A are the manager's and the administrator's; the floor has its own document, the *Resin Pouring Operator Guide*, written to be read at the pump.

It covers more ground than the plant needs today, and the marker on each section says which is which: **In service** runs now; **Optional** is finished and switched off until somebody wants it; **Not in service** was built to find out whether an approach works, with nothing depending on it. **Part One is what is being proposed** — the rest is here so the capability is known, not because it is being asked for.

# A production record for resin pouring

Resin pouring has never had a production tracking system. Counts, downtime, changeovers and cartridge lots lived on paper and in people's heads, which means the plant has been running without a record it could query, an hourly pace it could see, or any way to answer a quality question after the fact.

This is that record. A web application opened in a browser — on an operator's phone at the pump, on a desk in the office, and on the floor display — backed by a PostgreSQL database. Operators log the hour as it happens, at the pump. That log is the whole input: the figures on the wall display, in the analytics charts and in the exports are not separate reports, they are the same rows read different ways.

Nothing here asks a manager to enter anything, and nothing asks the plant to find somebody to administer it: as a logging system a manager holds the console too, so a PIN reset or a new pump does not wait on a second person. The system installs with work orders switched off — a plant that wants to dispatch runs to stations and track them against a target turns that on, and the screens for it appear; a plant that does not is a log, and the operator terminal never mentions runs at all. Everything in Part Two of this handbook works the same way: a setting, answering a question somebody asked first, not a step in getting the record.

**1**

screen an operator uses, on the phone already in their pocket

**0**

entries required from a manager for the record to be kept

**3**

roles, each seeing only what its job needs

**2**

shifts, a plant setting rather than an assumption in the code

**18**

database tables of production, quality and configuration

**1,420**

automated checks run before a release, including a browser driven through a shift

## What it records

- Every unit poured, attached to an operator, a station, a resin, a lot number and a timestamp.
- A lot check against each pouring log — every container format — including the ones that failed.
- Downtime by reason and duration, and scrap split between empty and filled containers.
- Start-of-shift, changeover and spill photo audits, filed against the station.
- Pre-shift validation per operator, per shift, per pump station.
- Reactor volume, reconciled against what has been logged rather than overwritten.

### SCOPE

The system is in service for resin pouring. Pack-out logging and machine integration are built but not in use; both are described in Appendix D rather than here, because neither has been committed to.

## Questions that had no answer before

A record is only worth keeping if somebody can ask it something. These are the questions resin pouring could not answer, and what answers them now. None of them needs an entry from anybody but the operator at the pump.

### QUALITY

#### Which containers came out of this lot?

Every pouring log carries the lot number the operator read off the container in front of them, checked against what was expected, with the operator, the station and the time on it — including the checks that failed. A complaint six weeks old becomes a query rather than a reconstruction.

### OUTPUT

#### What did we pour, and when?

Hourly, by operator, station, resin and container format, from the day it was switched on. Exportable for anybody who needs the numbers somewhere else, in the shape they are already kept.

### LOST TIME

#### Where did the hours go?

Downtime logged against a reason and a duration while it is happening rather than remembered afterwards, and scrap split between empty and filled containers. Both ranked, so the biggest one is visible instead of the loudest one.

### PACE

#### How is the shift running, right now?

The floor display shows the team its own rate against the expected figure while there is still time to do something about it — built from the same hourly entries and nothing else.

### STOCK

#### How much is left in that vessel?

Worked out from what has been poured out of it since it was last filled. Nobody dips a tank and nobody keys a number in for this to be right.

### DISCIPLINE

#### Is the lot check actually being done?

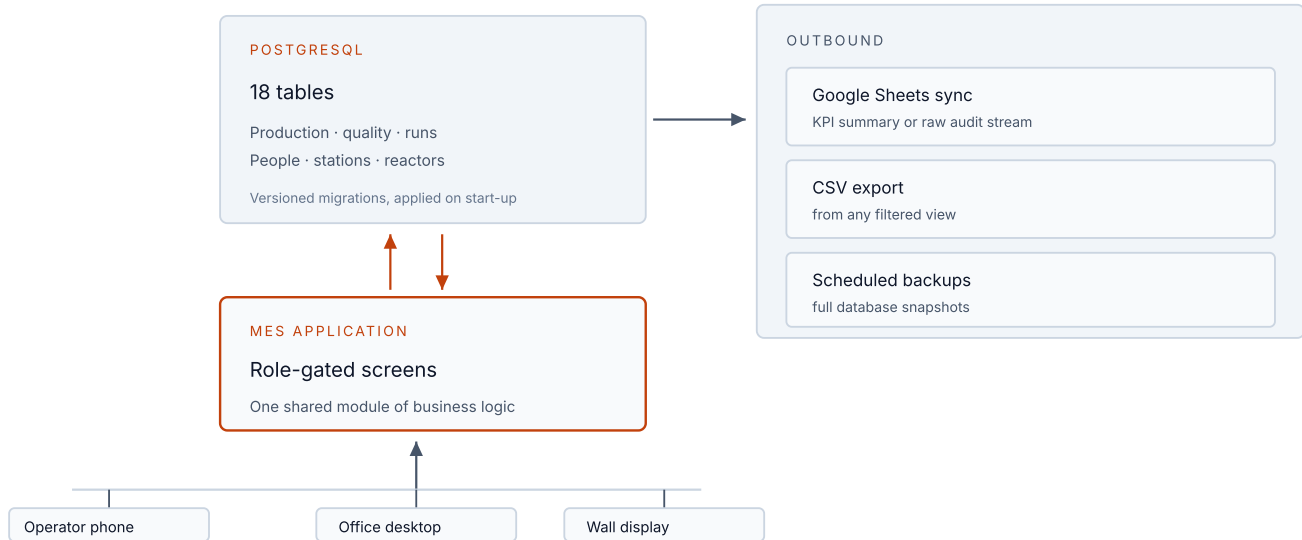
Full checks against confirmation taps, per operator and per station. A station showing almost no full checks is taking the fast path every hour, and that shows.

### WHAT IT COSTS THE FLOOR TO GET THIS

One screen, on the phone the operator already carries, filled in once an hour — no new hardware, no scanner, no terminal to buy. And **nothing entered by anybody upstream**: a manager who never opens it still gets the whole record, because the record is the operator's log and there is no second copy of it anywhere.

# One database, many windows

Nothing on the floor keeps its own copy of the numbers.



## Schema changes

Structural changes ship as numbered migrations that apply themselves on start-up, rather than SQL run by hand. Logs carry foreign keys to the operator and station as well as the legacy name text, so renaming someone does not split their history.

## Where the logic lives

Business logic sits in one module the whole application shares. Every screen that shows a figure reads it through the same functions, so a change to how yield is calculated applies everywhere at once.

## What happens when an operator submits a log

- 1 Checked at the station**  
 The lot on the container is compared against the run before the form accepts the entry.
- 2 One transaction**  
 The log and its verification record are written together; neither can exist without the other.
- 3 The run re-counts**  
 Progress is recomputed from the logs matching its station, resin, format and lot.
- 4 Everything else follows**  
 Display, analytics and exports all read those same rows.

# The operator terminal

One screen, sized for a gloved thumb on a phone: stacked inputs, large targets, no navigation to get lost in. Operators open it on their own handsets rather than at a shared terminal.

The screenshot shows a mobile application interface for an operator terminal. On the left is a dark sidebar with the user's name 'Ana Ruiz', role 'OPERATOR', and shift 'Shift 1'. It contains navigation buttons for 'Live SCADA', 'Workstation', and 'Live Reactors', along with an 'Account & Preferences' dropdown and a 'LOG OUT & CLEAR DEVICE' button. The main content area is titled '1. Station & Material Setup' and includes three dropdown menus for 'Pump Station' (New Pump #1), 'Container Format' (V2 (1L Cartridge)), and 'Resin Formulation' (Bioresin XZ9). Below these is a visual representation of a resin cartridge with a target weight of 1050.0g and a range of 1040-1060g. The second section, '2. Cartridge Lot Verification', instructs the user to turn the cartridge over and enter the lot number from the bottom label. A text input field shows the lot number '241180742'.

**The hourly log.** Station and material, the lot read off the container, then the counts. This is the screen in full — there is nothing above it and nothing after the submit button.

## TAB 1

### Log hourly pouring

Station, container format, resin, verified lot, good units and both scrap categories, with the target fill weight and tolerance for the selected resin shown alongside.

## TAB 2

### Log station downtime

Reason from the plant's configurable list, duration and notes. Feeds the downtime ranking used in analytics.

## TAB 3

### Cleanliness & photo audit

Start of shift, end of shift, line transfer or spill, with a camera capture or upload, filed against the station.

## TAB 4

### Manager comms

A direct message thread with the plant lead, inside the terminal.

## ALWAYS VISIBLE

Units poured, scrap and quality yield for that operator sit at the top of the screen and update as they log. It is the same figure management sees; there is no separate scoring.

# Lot verification

Lot labels get mixed up at the pumps: a pallet of pre-labelled empties is staged that belongs to a different lot, and resin goes into it. An earlier build of this form auto-filled the run's lot and printed it on screen, which meant an hour could be logged without a cartridge ever being turned over — the check was answering its own question.

**2. Cartridge Lot Verification**

Expected lot for this run: \*\*\*\*\* hidden on purpose — read the cartridge, not the screen

Turn the cartridge over. The lot label on the bottom reads L- followed by the lot. Type it exactly as printed — spacing, case and the prefix don't matter.

L- — lot on the cartridge bottom label

2411A0742

**The check.** The lot the run expects is masked. The operator turns the container over and types the L- code from the lot label on its bottom. Spacing, case and the prefix are normalised, so a code typed verbatim matches a run lot entered bare.

**STOP — DO NOT POUR.** This cartridge is not from the lot assigned to your run. Set it aside and get your lead.

Pulled the cartridge instead? Log the catch — it belongs in the record.

**WRONG CARTRIDGE — PULLED IT, NOTHING POURED**

Logging it anyway? Say what happened.

Choose an option

Details (required)

**When it doesn't match.** Submission is refused until the cartridge is pulled — recorded as a catch, with no production logged — or a cause, details and a photograph of the label are supplied. The log then saves flagged, carrying the lot physically in the container.

## FREQUENCY

### Full check, then a confirm

The full check runs on the first log of a shift, on any change of run, lot, resin or station, after four hours, and on every tenth log. Otherwise a single confirmation tap.

## COST

### Photo only on a mismatch

A clean check is one field and one tap. The photograph is required only when logging against a flag — and deliberately not required to pull a cartridge, so the safe action is never the slower one.

## 2. Jug Lot Verification

Expected lot for this run: \*\*\*\*\* hidden on purpose — read the jug, not the screen

Turn the jug over. The lot label on the bottom reads L- followed by the lot. Type it exactly as printed — spacing, case and the prefix don't matter.

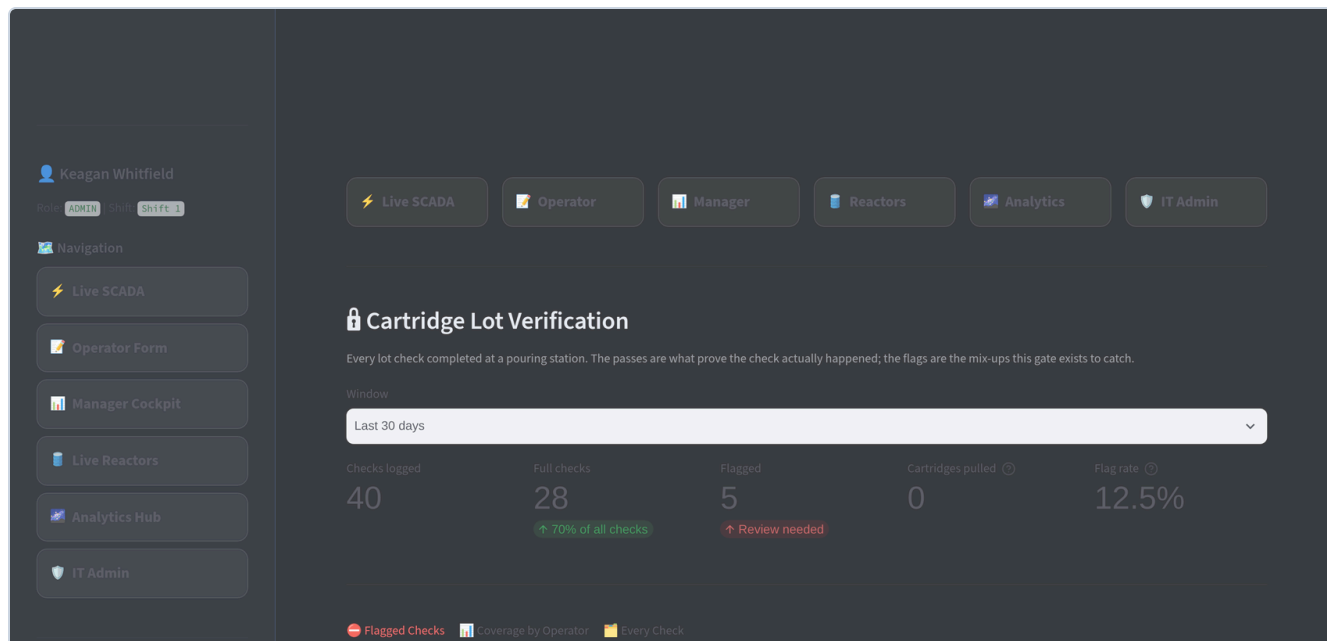
L- — lot on the jug bottom label

**The same check on a bulk jug.** RPS was the one exempt format, on the understanding that jugs carried nothing to read. They carry the same lot label a cartridge does, in the same place — so it is the same check, and only the noun changes with the format picked above.

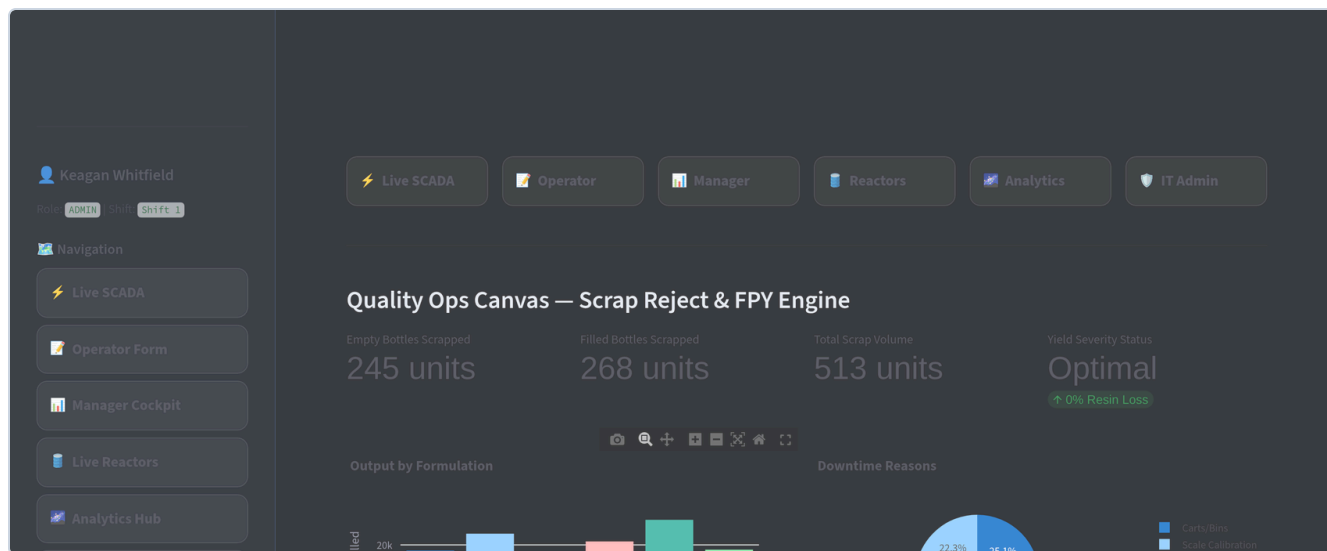
## DESIGN NOTE

A mismatch does not stop the line; it asks for an explanation, because a control that halts production is one people learn to work around. Every flag and every pulled container reaches the manager's review page.

# Review and scrap intelligence



**Lot Verification review.** Every check completed, pass or fail. The passes record that the check happened; the flags are the mix-ups. Coverage per operator shows who is completing full checks and who is using the confirmation tap.



**Scrap & Yield Intelligence.** First-pass yield against the configured target, with scrap split between empty and filled containers — a handling problem and a lost-resin problem respectively — and downtime reasons ranked alongside.

## PHOTO AUDITS

### Cleanliness gallery

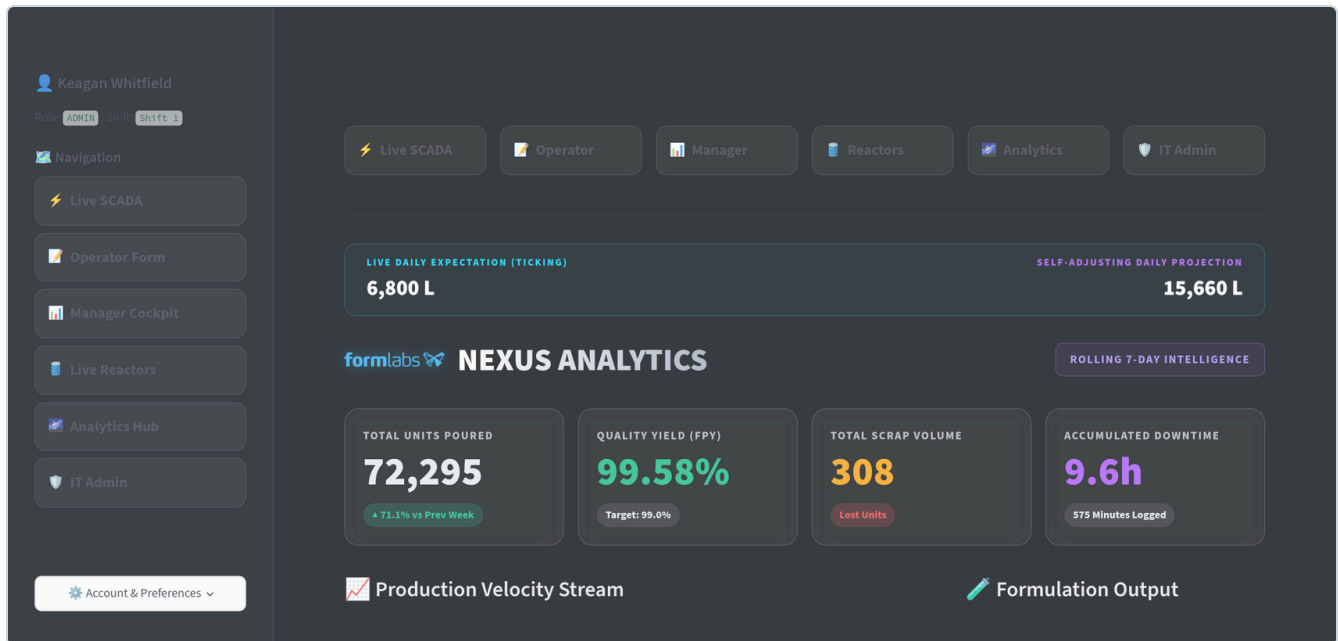
Start of shift, end of shift, line transfer and spill audits with photograph, station, operator and timestamp. Spills are counted separately.

## EXPORT

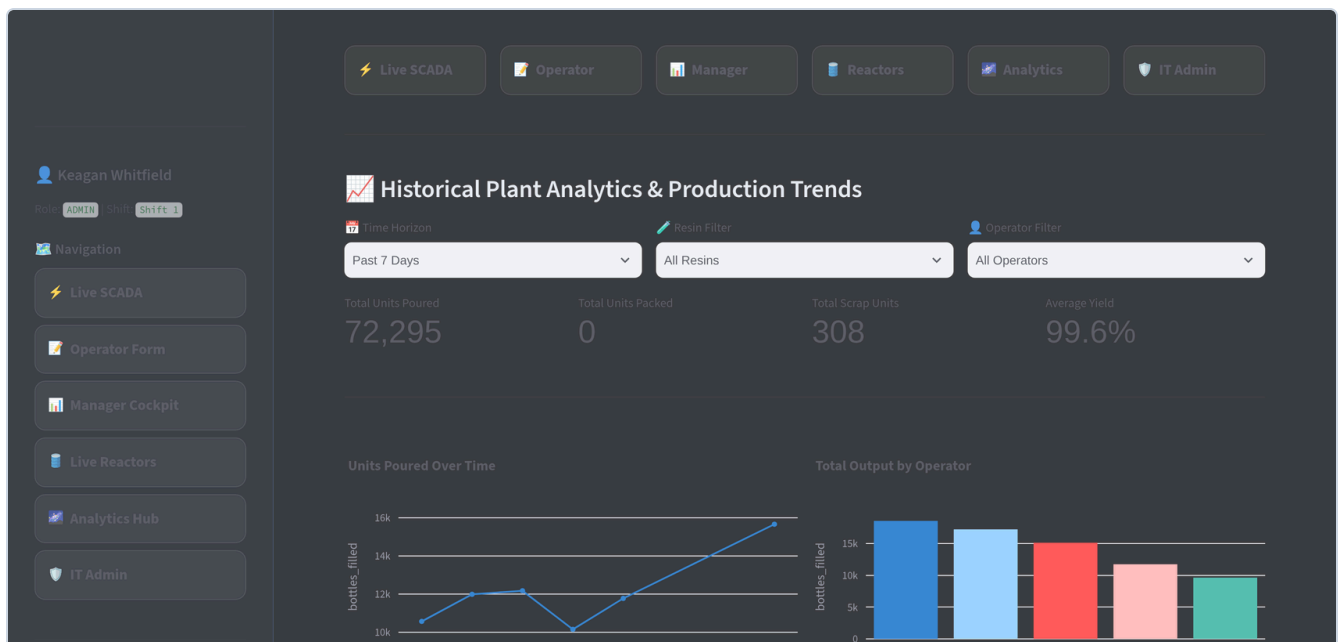
### CSV from the register

The verification register exports directly, so a quality question or customer audit can be answered without a database query.

# Trends



**Analytics Hub.** Output trend over the selected horizon, share of production by resin formulation, and downtime ranked by reason.



**Historical Production Trends.** The same data filtered by horizon, resin and individual operator, for month-to-date, year-to-date or any range.

## Reporting out

Getting the record out of the application, for somebody who is not going to open it. Both routes are manual and on demand: nothing is scheduled, nothing is emailed, and nothing leaves the plant unless a person asks it to.



**External reporting.** Pick the payload — an aggregated summary or the raw audit stream — the time horizon and the columns, then push. Internal-only fields are stripped before anything leaves the app.

### CLOUD EXPORT

#### Google Sheets sync

For the people who work in a spreadsheet and are not going to stop. The sheet is a copy, not a second system: it is written from the same rows and never read back. Each manager links their own spreadsheet rather than sharing one the installer set up, and a sheet stays private unless its owner shares it.

### SELF-SERVICE

#### Excel or CSV, straight down

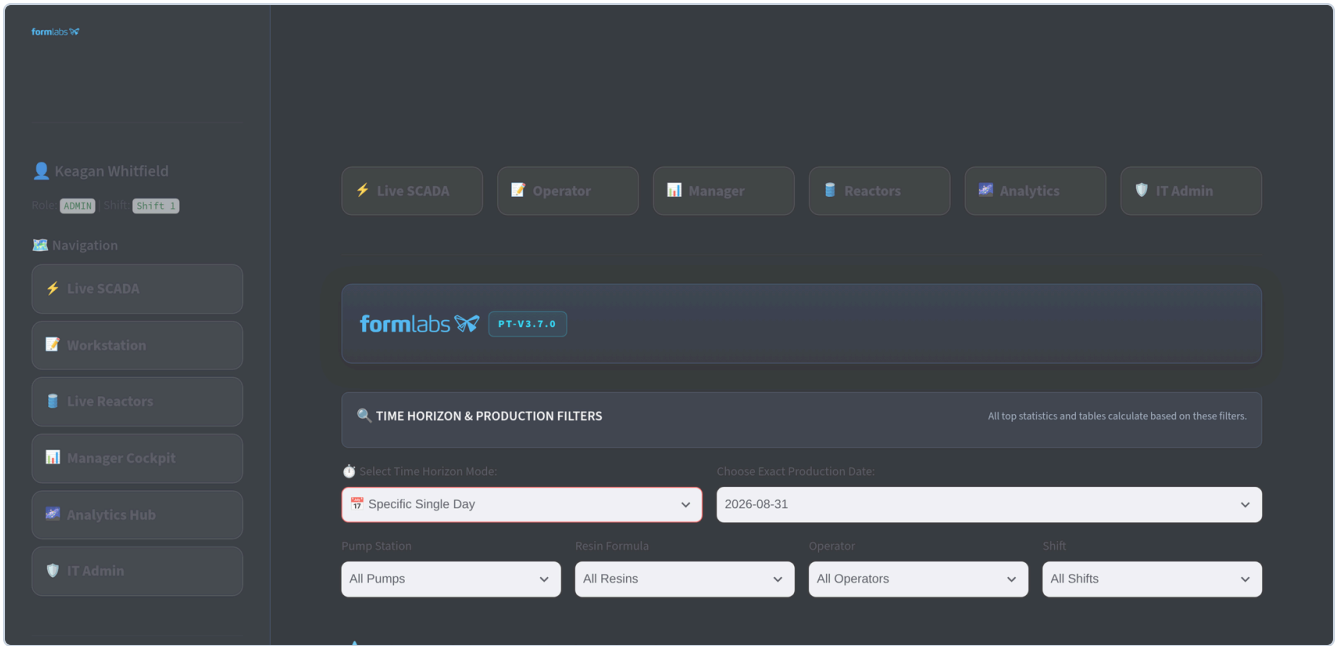
The same rows as a file — no account, nothing to publish, nobody's permission to ask for — opening straight into Sheets or Excel. This is the route that cannot fail: a company Google account usually sits under a policy forbidding the kind of connection a live sync needs, and no amount of correct setup gets past it. Filtered log views and the verification register export directly too, so finance, quality or an audit can be answered without a developer.

### REMOVED, AND WORTH SAYING WHY

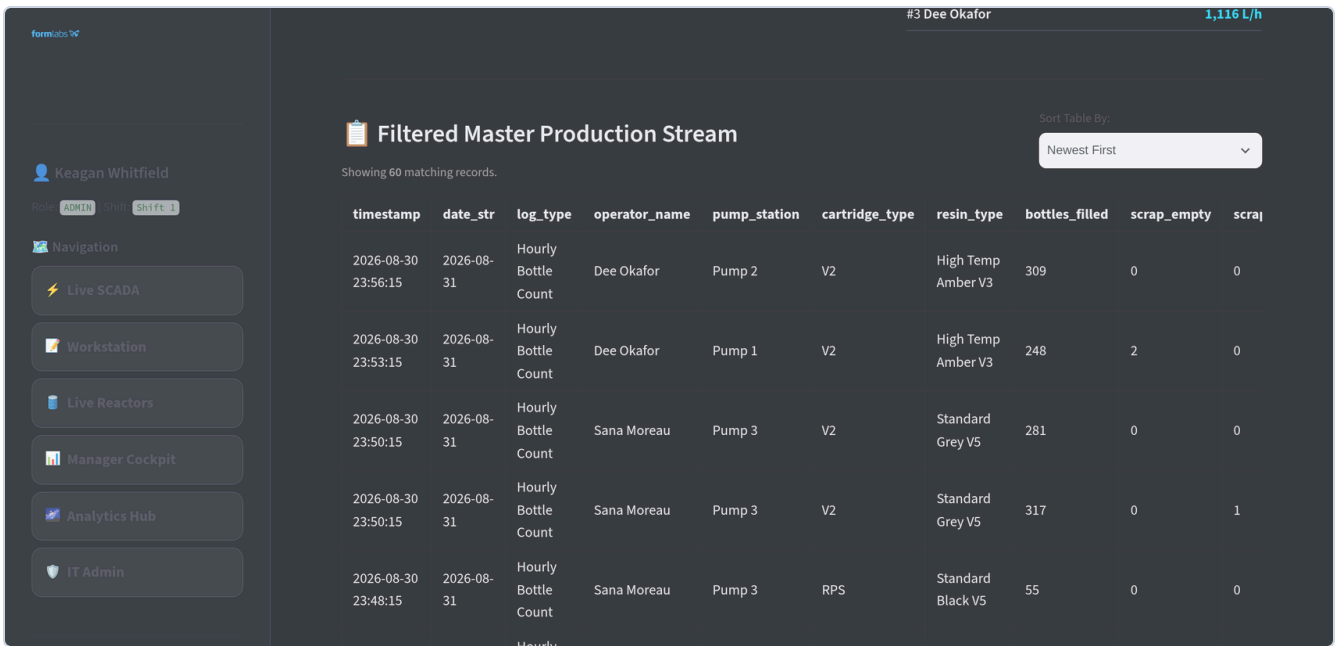
There was a shift handover here: four whole-day figures rendered as a PDF and emailed to a distribution list. Every one of them is on the landing screen, broken down by station, operator, resin and shift — which the report was not, since it filtered by date rather than by shift. The lead who arrives before the shift and opens the app was always going to see more than it could tell them, and it was the only feature in the application that needed an outside credential to work at all.

# The floor as it is now

The landing screen for every role. Filters across time horizon, station, resin, operator and shift recalculate every figure on the page.



**Live SCADA.** Volume output, resin mass poured (calculated from each formulation's master fill weight), run velocity against the configured target, and pouring yield.



**Below the fold.** Shift trajectory against pace, a pouring leaderboard, and the filtered production stream — the individual logs behind the figures above.

# Roles

Everyone signs into the same application and sees a different one. Access is checked on every page, so the difference is enforced rather than cosmetic.

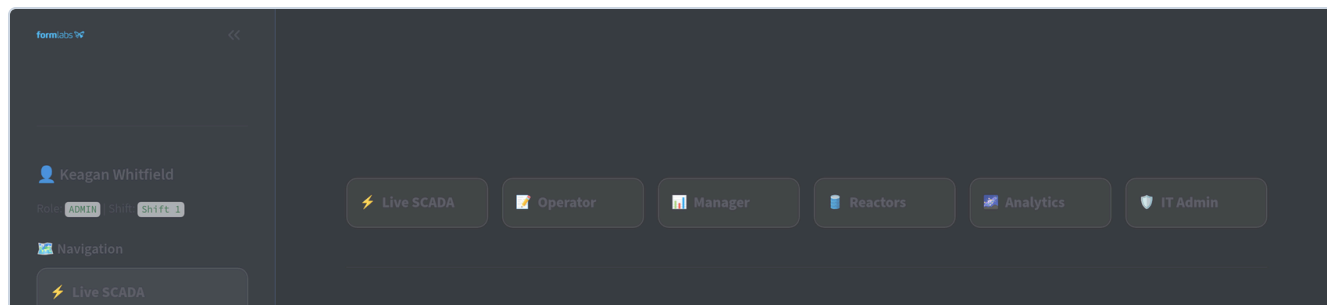
The column that moves is **Manager**. A plant running this as a logging system has no separate IT role — there is nobody to be it — so a manager is the administrator: accounts and PIN resets, equipment, plant settings, backups and cleanup. Switch to an execution system and those hand back to administrator accounts, which is what the shaded rows below mean.

| CAPABILITY                                    | OPERATOR | MANAGER      | ADMIN     |
|-----------------------------------------------|----------|--------------|-----------|
| Live SCADA dashboard                          | Yes      | Yes          | Yes       |
| Startup checklist & terminal unlock           | Required | n/a          | n/a       |
| Log hourly pouring & lot verification         | Yes      | Test mode    | Test mode |
| Log downtime & cleanliness audits             | Yes      | Yes          | Yes       |
| Message the plant lead                        | Yes      | Both ways    | Both ways |
| Live reactor fleet & reconciliation           | Yes      | Yes          | Yes       |
| See the lot a run expects                     | Masked   | Yes          | Yes       |
| Create & dispatch work orders                 | —        | Yes          | Yes       |
| Analytics hub & historical trends             | —        | Yes          | Yes       |
| Lot verification review & scrap intelligence  | —        | Yes          | Yes       |
| Google Sheets sync & CSV export               | —        | Yes          | Yes       |
| Floor roster & PIN resets                     | —        | Floor staff  | All users |
| Master resin specifications                   | —        | Yes          | Yes       |
| Log management & bulk cleanup                 | —        | Yes          | Yes       |
| Plant configuration, shift count & parameters | —        | Logging mode | Yes       |
| Record a check weight against an hourly log   | Optional | Optional     | Optional  |
| Undo your own last log, within two minutes    | Own only | Own only     | Own only  |
| Database backup & restore                     | —        | Logging mode | Yes       |
| Provision accounts & reset any PIN            | —        | Logging mode | Yes       |
| Choose how the plant runs this system         | —        | Logging mode | Yes       |

*Logging mode* marks the rows a manager holds only while the plant runs as a log. *Test mode* lets a manager or administrator log on behalf of an operator for training and troubleshooting; the entry is attributed to the operator selected and is visible in the log.

One thing the mode cannot do is strand a plant: an execution system restricts this console to administrator accounts, so the switch is refused while no such account exists, rather than locking everybody out of the screen that would switch it back.

# Running the system



**IT Administration.** Roster, the issue inbox, database utilities and global plant configuration.

## PEOPLE

### Accounts and roles

Provision staff, assign role and shift, set per-operator rate targets, reset PINs and unlock accounts. Managers can provision floor personnel without full IT access.

## PLANT

### Global parameters

**How many shifts the plant runs** — this one runs two — with start times, lengths and breaks; rate and yield targets; and the address of the pump form the operator terminal links out to.

## EQUIPMENT

### Stations, reactors, reasons

Pump stations, the reactor fleet and capacities, and the downtime reason list, so the vocabulary stays the plant's own.

## RECOVERY

### Backup and restore

A full snapshot is taken automatically once the newest one is a day old, the last fortnight is kept, and IT Admin says in one line how old the newest is — so a backup that has quietly stopped is visible rather than assumed. Each one records what the database held when it was taken, and a restore onto another machine is **checked against that row by row** before anybody is invited to start using it.

Quick Search by SKU, Resin Name, or Code

Showing: ALL (16 Resins)

| ID | FORMAT | SKU           | CODE     | RESIN NAME      | TARGET (G) | MIN (G) | MAX (G) | TARGET (KG) | MULTIPLIER | LIFETIME |
|----|--------|---------------|----------|-----------------|------------|---------|---------|-------------|------------|----------|
| 8  | RPS    | RS-R5-GPBK-05 | FLGPBK05 | Black V5        | 5000.0     | 4950.0  | 5050.0  | 5.0         | 1.0        | 24       |
| 6  | V1     | RS-C1-DRV2-01 | FLDRV201 | Draft V2        | 1120.0     | 1110.0  | 1130.0  | 1.12        | 1.0        | 24       |
| 11 | V1     | RS-C1-FL80-01 | FLFL8001 | Flexible 80A V1 | 1070.0     | 1060.0  | 1080.0  | 1.07        | 1.0        | 24       |
| 7  | V1     | RS-C1-TO15-01 | FLTO1501 | Tough 1500 V1   | 1070.0     | 1060.0  | 1080.0  | 1.07        | 1.0        | 24       |
| 16 | V2     | RS-C2-BIOX-01 | FLBIOX01 | Bioresin XZ9    | 1050.0     | 1040.0  | 1060.0  | 1.05        | 1.0        | 24       |

**Master Resin Specifications.** Target fill weight, tolerance band and shelf life — what every check weight is judged against, and what turns a bottle count into resin mass. Each formulation carries its colour from the plant's own master sheet.

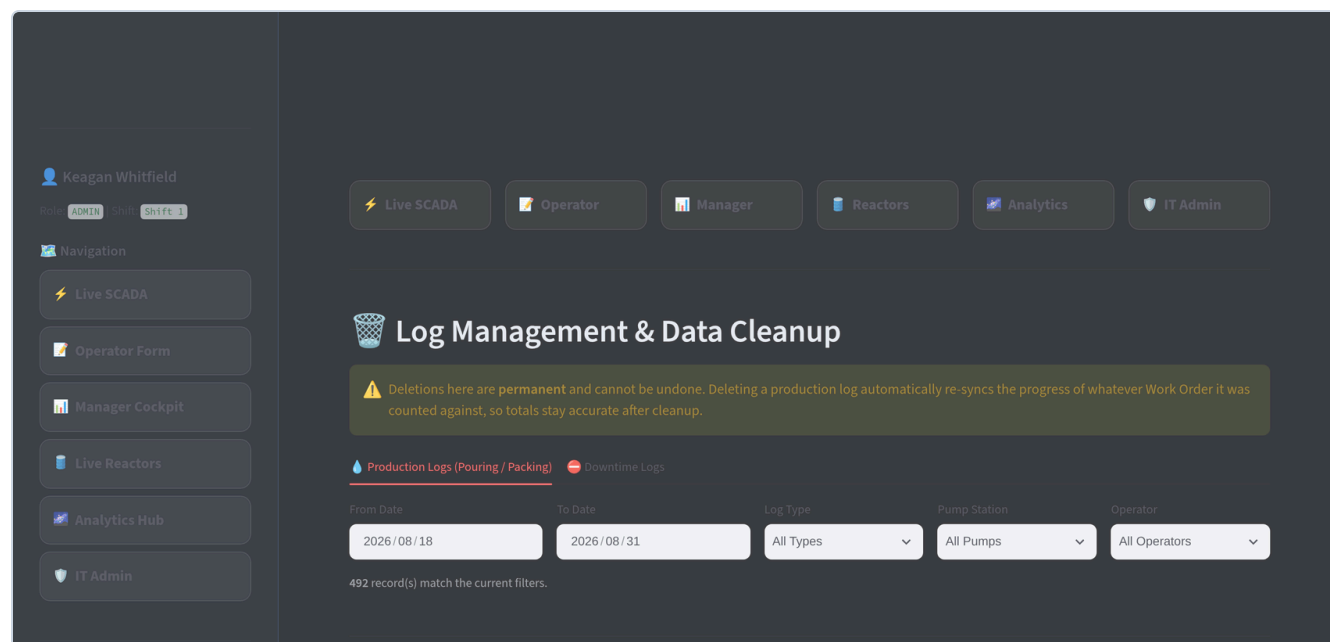
# Access control and data integrity

## Accounts

- **PINs are hashed.** Salted with bcrypt; the database holds no readable PIN, including for administrators.
- **Lockout.** Five failed attempts locks the account for fifteen minutes, with an instant IT unlock.
- **Sessions are tokens.** A signed token with a fixed lifetime backs "remember this device", revocable server-side.
- **Every page checks the role.** Enforced on the page itself, so typing a URL for a screen your role lacks returns access denied.
- **Output is escaped.** Values typed on the floor are escaped before rendering, so a note cannot inject markup or break its own card.

## The record

- **History survives a rename.** Logs carry foreign keys to the person and station as well as the name text.
- **Deletions re-sync.** Removing a log recalculates whatever work order it counted toward.
- **Bulk deletion is deliberately hard.** Disabled until the word DELETE is typed into a confirmation box.
- **Corrections are visible.** A plant-level correction is written as a labelled system adjustment, not by editing an operator's entries.
- **Credentials are kept out of the application itself.** Connection strings and passwords live in a configuration file on the machine, not written into the program.



**Log Management & Cleanup.** Filter production and downtime logs by date, station, resin or operator, review, and delete with the work-order re-sync handled automatically.

### WHEN NOTHING IS ARRIVING AT ALL

The failure worth planning for is not corruption, it is silence. If this machine reboots and the database does not come back, or the application is closed and nobody reopens it, every figure on every screen carries on looking entirely normal — because every figure is computed from the log, and the last good hour is still the last good hour. So the system watches its own intake: after roughly three hours with nothing logged *during a running shift*, the main screen and the floor display both say so plainly. Quiet outside a shift is not treated as a fault, because a warning that shouts all night is one people stop reading. Nor is a day the plant does not run: the days of the week this plant works are set once in the admin console, and a shift counts by the day it starts, so a Friday night shift is still watched into Saturday morning. An alarm that cries wolf every weekend is worse than no alarm at all — by Monday nobody reads it, and the one that means something looks exactly like the fifty that did not.

## Vessel levels and the floor display

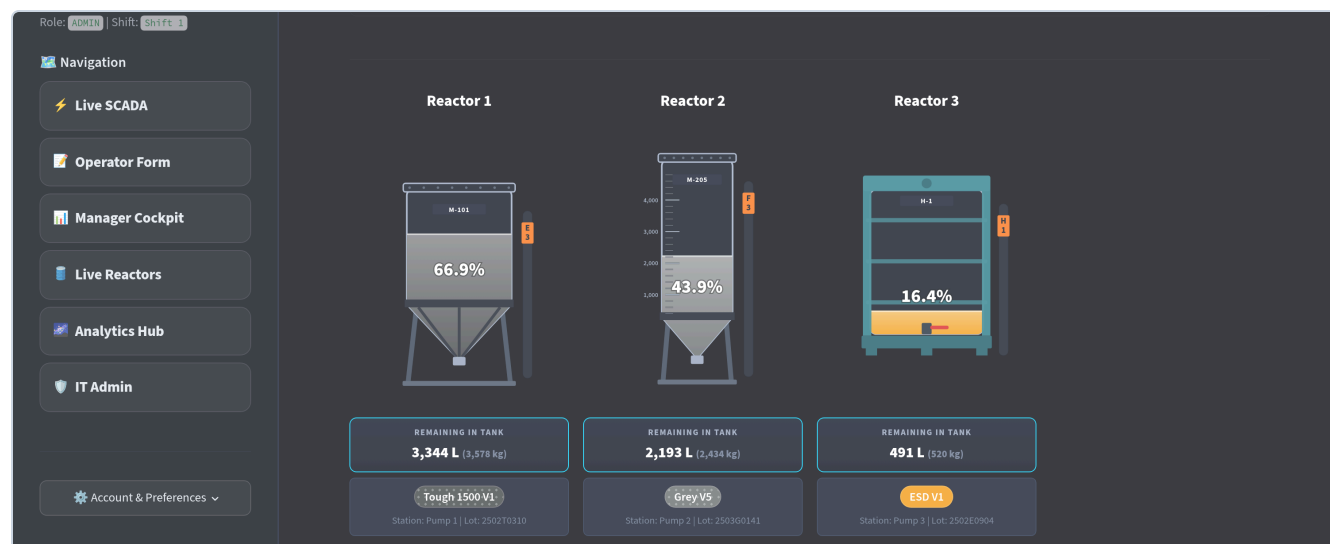
How much is left in each supply vessel, worked out from the same hourly logs as everything else — and the same figures on a wall screen the floor can read without opening anything.

A vessel's level is not a number anybody types in and it is not read off an instrument. It is the vessel's capacity less what the record says has been drawn from it since it was last filled, and both halves of that come from the hourly log the operator is already filling in. Each log carries the lot number the operator read off the container, so a new lot at that station is the vessel having been refilled; and each log carries the container format it was poured in, so a bulk jug and a cartridge are counted at their own volumes rather than one assumed for all. Nothing has to be entered anywhere else for the levels to be right.

Not everything leaving a vessel is a counted container. An amount decanted into a drum or a tote is entered as the amount — in litres or in kilograms — and converted using the density already implied by the resin's own fill-weight spec, so no new figure has to be looked after. An amount larger than the vessel could have given out is refused at the keyboard, because a typed volume is not self-checking the way a container count is: nobody types 2,500 cartridges and believes it, but 1800 litres instead of 180 looks perfectly ordinary and only shows up an hour later as an empty vessel on the wall.

The consequence worth stating plainly: the level does not depend on work orders. It reads identically whether or not the plant is dispatching runs.

When a gauge disagrees with the calculated figure — a transfer nobody logged, a partial fill — anyone at the vessel can correct it from the operator terminal, either by the percentage on the sight glass or by the exact litres. The correction is written into the record as a dated, attributed adjustment rather than quietly replacing what was there, so the disagreement stays visible and the level goes on being derived from the record afterwards.



**Live Reactor Fleet.** Each vessel drawn as its own kind, with its asset tag, bay marker, resin, station, lot and remaining volume in litres and in kilograms.

### WHEN A VESSEL SITS AT EMPTY

A vessel that reads empty while pouring continues from it is telling you something specific: more has been drawn on the current lot than the vessel holds, which means it was refilled and the lot on the log did not change with it — the same drum number read off a second container, or a top-up nobody recorded. The

## Reading a vessel off the screen

The picture is drawn to be checked against the vessel it describes, because a screen that can be checked at a glance gets checked.

Each vessel is drawn as the kind of thing it physically is — a tall bulk reactor, a squat cone mixer, a caged tote — because on this floor those are three different objects and their capacities do not tell them apart. Every fabricated vessel here has a cone bottom in a steel frame; only the totes are flat, and what separates the two fabricated ones is their proportions and whether litres are painted on them. Each carries its stencilled asset tag and the orange bay marker from the bollard beside it, so the vessel on the screen and the vessel in the aisle have the same name. And where the real vessel has litre graduations, the drawing has them too, with the surface landing on the mark somebody standing there would read it against: a vessel showing 2,193 L sits just above its 2,000 line, on the screen and in the aisle.

The liquid is drawn in the resin's own colour, on the same reasoning as the coloured labels everywhere else — from across a room, colour is what identifies a formulation, and this wall of vessels is meant to be read from across a room. A vessel with nothing assigned to it reads **IDLE** rather than nought percent, which is a different statement and the true one.

### The three kinds, and what a manager sets

#### TALL BULK REACTOR

##### Tall and narrow

Bolted top flange, cone bottom in a steel frame, and litre marks painted up the barrel. Drawn with its scale.

#### SQUAT CONE MIXER

##### Short and wide

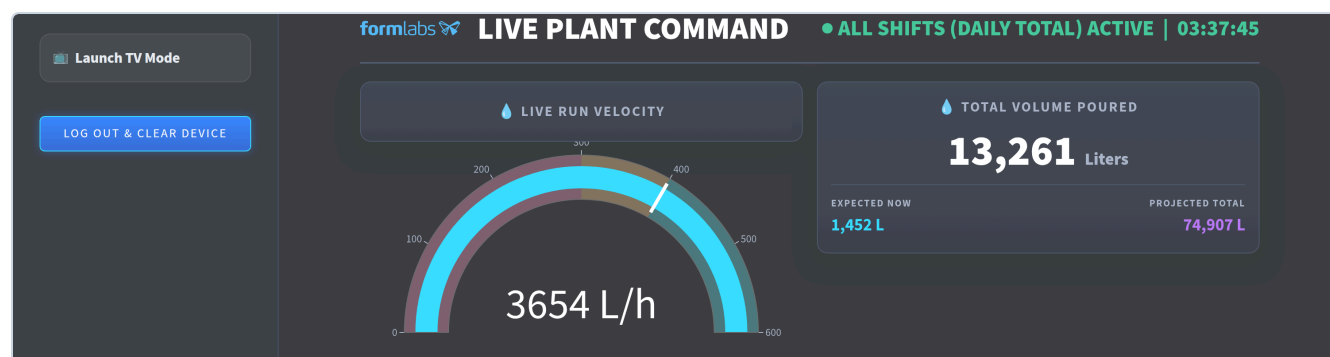
The same construction in different proportions, with ribs down the cone. Drawn without a scale, because the real one has none.

#### IBC TOTE

##### Caged bottle on a pallet

Not a fabricated vessel, and the only flat-bottomed one. Moulded cage, pallet feet, ball valve at the front. Around 1,000 L.

Each vessel carries three optional settings under *Manage Permanent Reactor Fleet*: which of the three it is, its asset tag, and its bay marker. The kind starts from the capacity, which is right for most of a fleet and wrong for the interesting ones — so it is a starting point a manager corrects, not a rule the drawing keeps re-deriving.

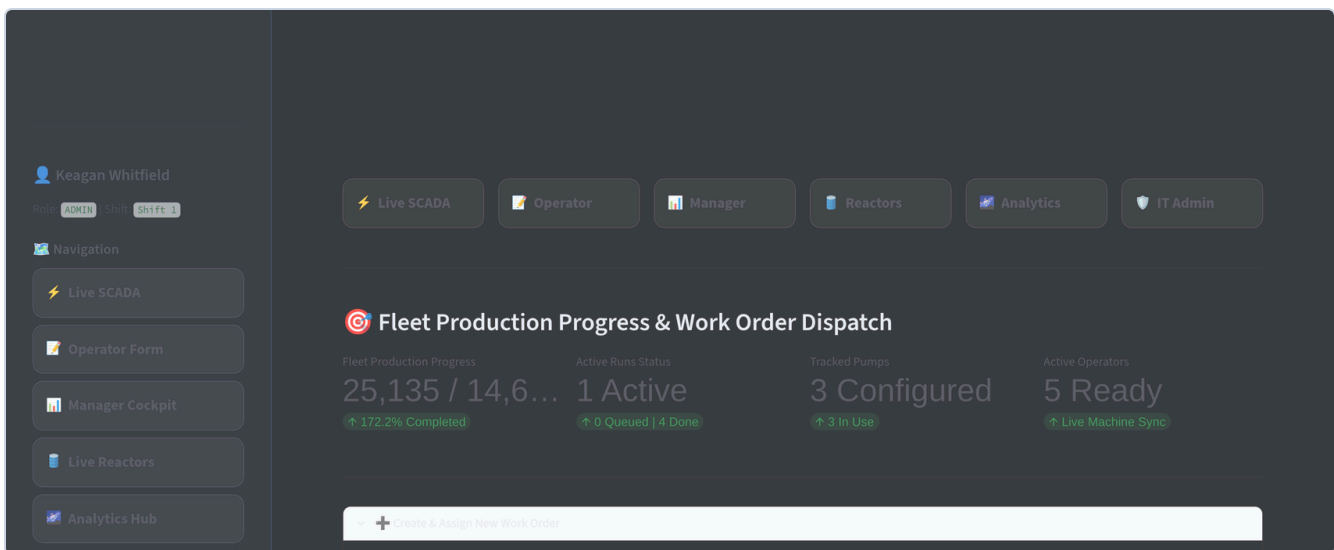


**Floor display mode.** The same record on a wall screen — the active shift, or the whole plant day, refreshing itself.

## Work orders

Off unless a plant turns it on, and the operator terminal in Part One is complete without it. This page is what becomes available if somebody wants work dispatched to stations and tracked against a target.

A manager creates a work order against a reactor, resin, container format and lot. Hourly logs submitted at that station are matched to it automatically, and the lot check gains something to compare against instead of only recording what was read.



**Work Orders & Assigned Runs.** Fleet-wide progress across active runs, with status, assigned operator, pump station and lot. Completed orders move to a separate history tab.

### DISPATCH

#### Create and assign

Reactor, size, resin, container format, target units, station and lot number, dispatched to a station rather than to one named person.

### COUNTING

#### Progress from logs

A run's count is recomputed from the production logs that match it, so deleting a bad entry re-syncs the order instead of leaving it overstated.

### SHARED STATIONS

#### Several operators, one run

Progress follows the pump station, so any number of operators can work the same run without a duplicate assignment being created for each.

### CLOSEOUT

#### Finish on the real number

A run can be completed on a custom final total when the physical count differs, and the reactor it drew from is released.

# Fill weight and giveaway

The system has always known what every cartridge is supposed to weigh. Until now it never recorded what one actually weighed, so it could say how many units were poured but not how much resin went into them.

## 16

resin specifications, each with its own target and tolerance window

## 1

optional reading an hour — never mandatory, never pre-filled

## 4 kg

given away per thousand cartridges by a pump running 4 g heavy

## Why a passing check is not the same as a right weight

A pump set up to stay safely clear of the low limit sits *above* target on every cartridge it fills. Every one of those passes. Every one carries resin the customer did not pay for. On a 1110 g target each gram over is about 0.09% of the fill — small on one cartridge, and 4 kg per thousand at four grams heavy. Four of the sixteen specifications make it worse by being lopsided: **Grey V5** and **Clear V5** allow 10 g under target and only 5 g over, so the safe place to sit is nearer the ceiling. The giveaway does not depend on that, though — it happens on a symmetric window too. It was invisible because the target was displayed and never compared against anything.

**3. Production Output**

☒ Good Units / Containers Filled

- +

Scrap Empty
 
- +

Scrap Filled
 
- +

Check weight (g) — optional ?

✖ - +

☒ In band — 4.0 g over target.

Process Observations / Notes

e.g. Target fill weight nominal...

**One box on the hourly form.** The reading is judged against that resin's own window as it is typed. Here 1114 g on a 1110 g target is inside spec and four grams heavy — a cartridge that passes and still gives resin away.

### OPTIONAL BY DESIGN

#### It can never block a log

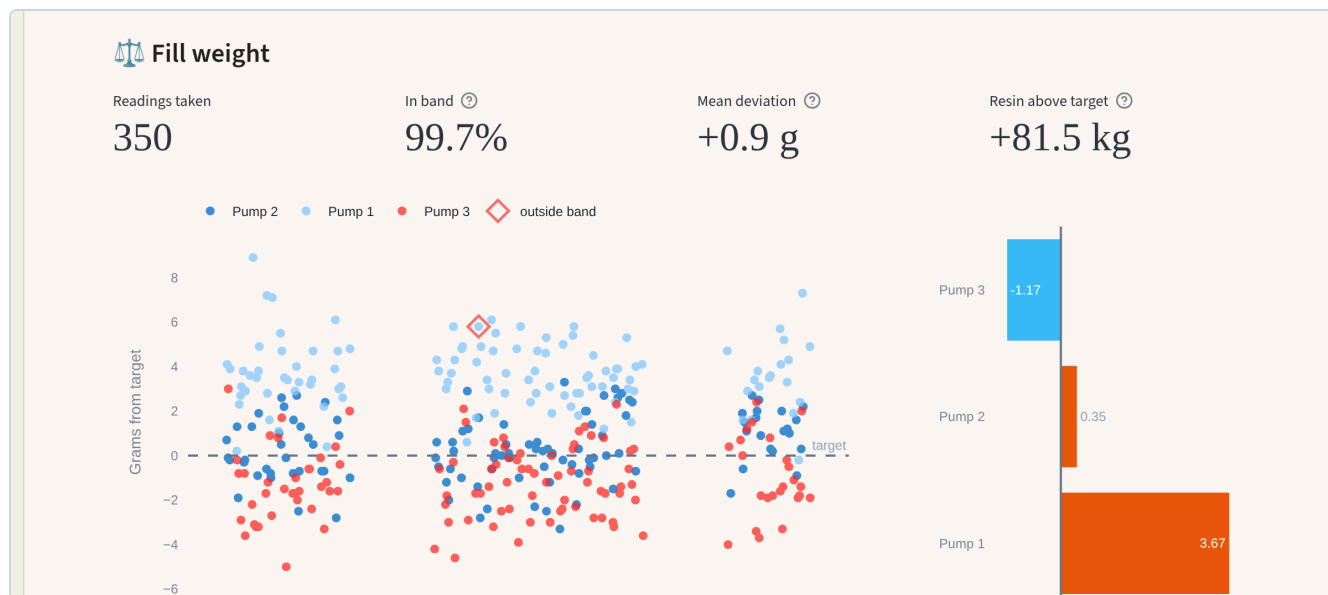
The lot gate stops the line because a wrong lot is a defect. A heavy cartridge is information. Make a weight mandatory and within a week it is the target typed from memory on every log — and a column full of **1110** is worse than an empty one, because it looks like data. Sparse and honest beats dense and invented.

### NEVER PRE-FILLED

#### The box starts empty

For the same reason the run card had to stop printing the lot the check was hiding: a field that already contains the right-looking answer gets accepted, not measured.

## What the readings add up to



**Fill weight, across a fortnight of pouring.** Deviation from target rather than absolute grams, so resins with different targets can share an axis; zero is target and out-of-band readings carry their own marker. On the right, mean deviation per pump — one running heavy, one light, one close to nominal. That is the whole argument in one chart: every point above the dashed line passed its check.

### ONE READING, NOT THREE

#### Between hours, not within a minute

Three cartridges weighed in the same minute mostly measure the scale. The variation worth catching is hour to hour, so one reading stands as a sample of its hour — and each is weighted by the units logged beside it, which is what turns “four grams heavy” into kilograms.

### JUDGED ONCE

#### The verdict is stored, not recalculated

A reading is compared against the window in force when it was taken, and that verdict is kept. Editing a specification later changes what happens next; it cannot retrospectively re-judge readings somebody already took.

### ONE NUMBER THAT HAS TO STAY HONEST

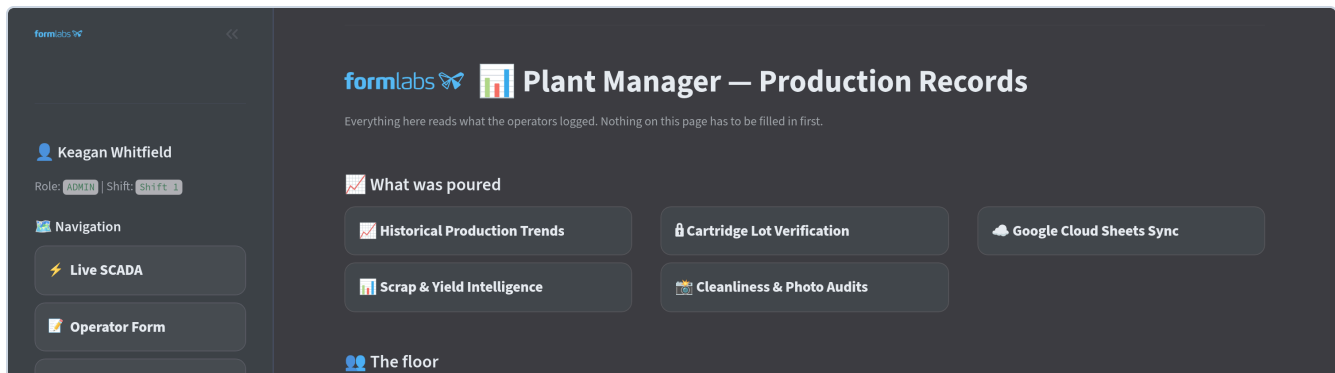
The headline share of readings inside their window never rounds a near-miss up to a clean 100%. 349 of 350 is 99.7%, and printing that as 100% directly above a chart in which the one exception is plainly visible costs the reader's trust in every other figure on the page — and that one exception is the reason to look.

### WHERE THIS LEADS

The machine-integration gateway described in Appendix D already carries a serial adapter written for exactly these bench scales. If they have a serial output the weight arrives with no operator effort at all — and these manual readings are what that automated capture would have to be validated against, since checking it needs known-good numbers to check against.

## Dispatching the day

This page applies only where work orders are switched on. Where they are not, a manager enters nothing at all — the next page, reading the floor and closing the day, is the whole of the role.



**Manager Cockpit.** Each management module on its own screen rather than one page trying to do everything.

- 1 Create the work orders**  
Reactor and size, resin, container format, target units, pump station and lot number. Assign an operator for accountability, but note progress follows the *station* — anyone working that pump contributes to the same run.
- 2 Enter the lot exactly as printed**  
The most important field on the form: it is what the operator's cartridge check is compared against. Enter the number as printed after **L -** on the base. Spacing, case and the prefix are normalised, but a wrong number cannot be.
- 3 Watch the fleet, not the logs**  
The work order page rolls up fleet progress, active and queued counts, tracked pumps and operators ready. Drill into a run only when its progress stalls.

### IF A PROGRESS BAR IS NOT MOVING

Almost always one of three things: the operator's station does not match the run's, the resin or container format differs, or the lot on the run does not match what is staged. The verification review shows which.

## Reading the floor, closing the day

### DAILY

#### Flagged lot checks

Mismatches poured anyway and cartridges pulled, each with the photograph, both lot numbers and the operator's reason. A mismatch that is really a formatting difference means the run's lot needs correcting, not the operator.

### DAILY

#### Scrap and downtime

Yield against target, the empty-versus-filled split, and downtime by reason. Empty scrap is a handling problem; filled scrap is lost resin and considerably more expensive.

### WEEKLY

#### Verification coverage

Full checks versus confirmation taps per operator. A station showing almost no full checks is one taking the fast path every hour.

### WEEKLY

#### Trends

Output trend, share by resin, and per-operator history. The historical page covers anything beyond the current fortnight.

4

#### Close out finished runs

Complete a run on its real final count when the physical number differs. The correction is booked as a labelled system adjustment and the reactor is released.

5

#### Answer the floor

The communications page holds every operator's thread. A question left unanswered there gets solved badly on the floor instead.

# Keeping it running

## BEFORE TRUSTING A MACHINE WITH THE PLANT

**Check\_This\_PC.bat** answers one question — is this PC ready — and when the answer is no it prints exactly what to type. It confirms the database is reachable and its schema current, that the backup tools are present *and new enough to read this server*, that there is disk space and a free port, and whether the firewall will let the handsets in. It takes a real backup while it is there, so backups are known to work rather than assumed. Then it prints the address to give the phones — the machine's name and its number, with the name marked, because a number stops working when the machine rejoins the network and every bookmark made from it dies with it.

In a plant running this as a logging system this page is the **manager's** job, not a separate person's — there is no separate IT role to hold it. The work is the same either way; only who does it changes.

### ON JOINING

#### Provision the account

Create the user with the correct role and shift before their first shift. Role decides which screens exist for them; shift decides which checklist and pace target apply.

### ON LEAVING

#### Remove access, keep history

Deleting a user removes access. Production history stays attached through the foreign key on every log.

### WEEKLY

#### Verify a backup

Take a snapshot and confirm it lands. A backup that has never been restored is a hypothesis; the restore path is on the same page.

### PER RELEASE

#### Let migrations run

Schema changes apply on start-up. After deploying, start the app once and confirm it comes up before the shift arrives.

### SEASONALLY

#### Review plant parameters

Shift times, lengths, break minutes, rate and yield targets. Every pace indicator and yield judgement is measured against these.

### MONTHLY

#### Clear the suggestions inbox

Staff file requests, bugs and floor issues from inside the app. An inbox nobody empties is one nobody writes to.

### NOT THIS

Do not edit production rows directly in the database to fix a number. Use log management to remove the bad entry — it re-syncs the affected work order — or a reconciliation to book a variance. Both leave a visible record; a direct edit leaves the totals right and the story wrong.

## Readability where the work happens

A phone held in a gloved hand at arm's length, in a room lit for pouring resin, is a different reading problem from an office monitor. These are the settings that address it. All of them are per-user and remembered; none of them changes what is recorded.

### GLOVES

#### Glove mode

Nitrile still registers on a touchscreen, but precision drops, and the pouring form asks for number steppers and a lot field. One toggle pushes every target past the accessibility touch size and scales the type. It is remembered against the *terminal*, not the account — being gloved is a property of where you are standing, not of who you are.

### LIGHTING

#### Light and dark

A dark screen is unreadable under bright shop lighting and a light one is glare on a wall display at night, so both exist and the operator picks. The choice follows the account, so a shared terminal does not undo it.

### CONTRAST

#### Measured, not assumed

Text is checked against the accessibility floor of 4.5:1 rather than eyeballed. The check found real failures in themes already in use — card labels in the default, and again across the sidebar, widget labels and popovers, where the worst measured 1.01:1. All fixed, and an audit screen re-runs the measurement on demand, so it is something you can see rather than something that was checked once.

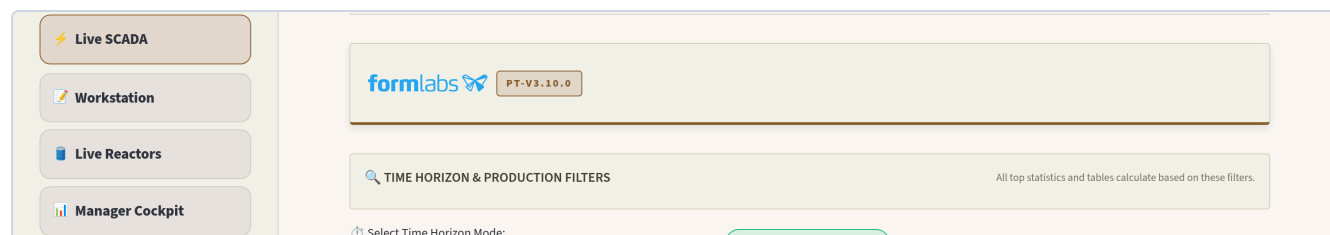
### NIGHTS

#### Automatic dimming

Derived from the plant's own configured shift times rather than a fixed hour, because “night” here means “not the day shift”. A gentle brightness and warmth reduction rather than a different set of colours. Photographs are exempt, so a lot label under review keeps its true colour.

## Resin colour, and why colour alone was not enough

Wherever a resin is named it carries the colour from the plant's own master sheet, so screen and sheet read as one document; a new formulation takes its family's colour from its name. But five pairs in that palette are the same colour to the eye, and a Clear/Rigid mix-up is exactly what the lot check exists to catch — so each chip also carries a **texture** from the resin family: Clear V4 and Clear V4.1 match, Clear and Rigid cannot. Texture survives greyscale printing and colour blindness, which a colour alone does not.



**The same screen, light.** Themes change presentation only — every figure, filter and control is identical, so a handbook page describes one screen regardless of how the reader has theirs set.

# How it is tested

Before a release the system is exercised end to end against a throwaway database built from nothing each time. It refuses to run against the production connection.

## 1,328

assertions across seventeen suites — workflow, interface, roles, colour, fill weight, shifts, vessel levels, vessel drawings, backups and the stopped record, machine readiness, the shift build, measured pours, export destinations and their round trip, links, boot paths

## 58

checks driving a real browser through an operator's shift, on a desktop and a phone

## 18

screens rendered and checked for errors on every pass

## What the simulation covers

All ten migrations are applied to an empty database and the result compared table by table against the application's own model definitions. A manager then creates three work orders; operators clear per-station checklists and pour — clean logs, a fast-path log, a mismatched cartridge, a cartridge pulled before pouring, a log with no run behind it, and a gated RPS jug — alongside downtime. Every figure the interface displays is then recomputed from the raw rows and compared against hand-figured expectations: poured, scrap, yield, per-operator totals, run progress, flag rate and variance.

## What that has caught

- **The run card was printing the lot the check was hiding.** An operator could have read the expected lot off the screen instead of the cartridge, which would have made the whole check decorative. Found because the interface test asserts the lot appears nowhere on the page.
- **Plant settings could never be saved.** The Admin Panel's save button passed a dictionary to a function expecting fifteen positional arguments, so shift times, rate targets and yield targets had never been changeable from the interface.
- **A content-delivery fetch could take the terminal down.** An unguarded network call for a decorative animation ran on every interaction; a floor PC that lost its internet would have hung the terminal before a control rendered.
- **Eight pages failed on a theme change,** and log cleanup crashed whenever a filter matched fewer than ten records — both long-standing.
- **A setting that saved and was never read back.** The number of shifts had a column, a migration, an admin field and a working save path — and the function every screen reads settings through never returned it, so all of them silently used the default. It happened to be the right answer for this plant, which is why nobody noticed.
- **Every light theme rendered its stylesheet as visible text.** Found in a browser after the tests had passed: an indented style block appended after other content is treated as a code listing and displayed rather than applied. The twenty-four older themes were never bitten because theirs comes first.
- **Every vessel on the wall would have drawn the first vessel's colour.** A gradient is referenced by an identifier, four vessels share one page, and duplicate identifiers all resolve to the first — so the second tank onwards would have shown the right level in the wrong resin's colour, on a screen whose whole point is being recognised by colour from a distance. Nothing would have looked broken.
- **Vessel levels only worked because work orders were switched on.** The level looked like a vessel feature and was not one. The work order list was quietly supplying both the point the vessel was last filled and the size of the containers coming out of it. With work orders off — which is how the system is proposed to run — every vessel summed every log ever taken at its station, emptied on the first day and stayed there, and counted a

## Experiments, not commitments

Two things exist in the codebase that are **not in use and not decided on**. They are described here so the capability is known, not because there is a plan attached to either. Both were built to find out whether the approach works.

### Pack-out logging

A packing role, a pack-out logging tab, and skid conversion using each resin's units-per-skid figure. Poured and packed totals are counted separately, and the gap between them reports as floor work-in-progress. A reconciliation flow books the variance between what was poured for a lot and the final packed count as a single labelled system adjustment.

It is switched off. Turning it on is a single plant setting, but nobody has decided whether pack-out should be logged this way, and the workflow has not been tested against how the line actually runs.

### Machine integration

A protocol-agnostic gateway for reading floor equipment directly, so counts and weights would not have to be typed. Each machine is a row in a registry and the polling logic sits behind one shared interface, with adapters written for Modbus TCP, Modbus RTU, OPC-UA, MQTT, Serial ASCII and HTTP. Readings would write through the same logging path the operator screen uses, so analytics and run tracking would need no changes. A registry screen for it is written — add a machine, point it at the pump station it sits on, map its raw tags — and is deliberately not linked from anywhere in the application, so it cannot be reached by an administrator who has not been told it is there.

#### None of it has been tested against real equipment.

The adapters were written to cover whatever the scales or pump controllers turn out to speak; which of them is correct is unknown until someone connects one. Until then the screen stays unlinked rather than deleted, so nothing is lost if the plant ever wants it.

The screenshot displays a web application interface for 'formulas'. On the left is a dark sidebar with a user profile for 'Keagan Whitfield' (Role: ADMIN, Shift: DAY 1) and navigation buttons for 'Live SCADA' and 'Workstation'. The main content area is titled '1. Packing Details' and contains a form with the following elements: a 'Location' dropdown set to 'End-of-Line / Pack-Out', a 'Container Format' dropdown set to 'V2 (1L Cartridge)', a 'Resin Formulation' dropdown set to 'Bioresin XZ9', and a highlighted 'Bioresin XZ9' button. At the bottom of the form, it says 'Batch Lot Number Being Packed'.

**Pack-out logging, currently disabled.** Container format, resin, lot and units packed, with the skid equivalent calculated as the count is typed.

#### READ THIS AS

Neither of these is a roadmap item or a commitment of dates. They are experiments that happen to be finished enough to demonstrate, and either could be abandoned without affecting anything in service.